EXERCISES

Queries in SQL.

<u>EXERCISE 1.</u> **(***Fundamentals of Database Systems*, Elmasri, Navathe, Addison-Wesley)

Specify the following queries on the COMPANY relational database schema.

(a) Retrieve the names of employees in department 5 who work more than 10 hours per week on the 'ProductX' project.
(b) List the names of employees who have a dependent with the same first name as themselves.
(c) Find the names of employees that are directly supervised by 'Franklin Wong'.
(d) For each project, list the project name and the total hours per week (by all employees) spent on that project.
(e) Retrieve the names of all employees who work on every project controlled by department 5.
(f) Retrieve the names of all employees who do not work on any project.
(f') Retrieve the names of all employees who do not work on every project
(g) For each department, retrieve the department name, and the average salary of employees working in that department.
(h) Retrieve the average salary of all female employees.
(i) Find the names and addresses of all employees who work on at least one project located in Houston but whose department has no location in Houston.
(j) List the last names of department managers who have no dependents.

<u>EXERCISE 2.</u> **(***Fundamentals of Database Systems*, Elmasri, Navathe, Addison-Wesley)

Specify the following queries on the COMPANY relational database schema.

(a) For each department whose average employee salary is more than $30000, retrieve the department name and the number of employees working for that department.
(b) Suppose we want the number of male employees in each department rather than all employees. The new query is: for each department whose average employee salary is more than $30000, retrieve the department name and the number of ***male*** employees working for that department.
Can we specify this query in SQL? Why or why not?

<u>EXERCISE 3.</u> **(***Fundamentals of Database Systems*, Elmasri, Navathe, Addison-Wesley)

Specify the following queries on the COMPANY relational database schema.

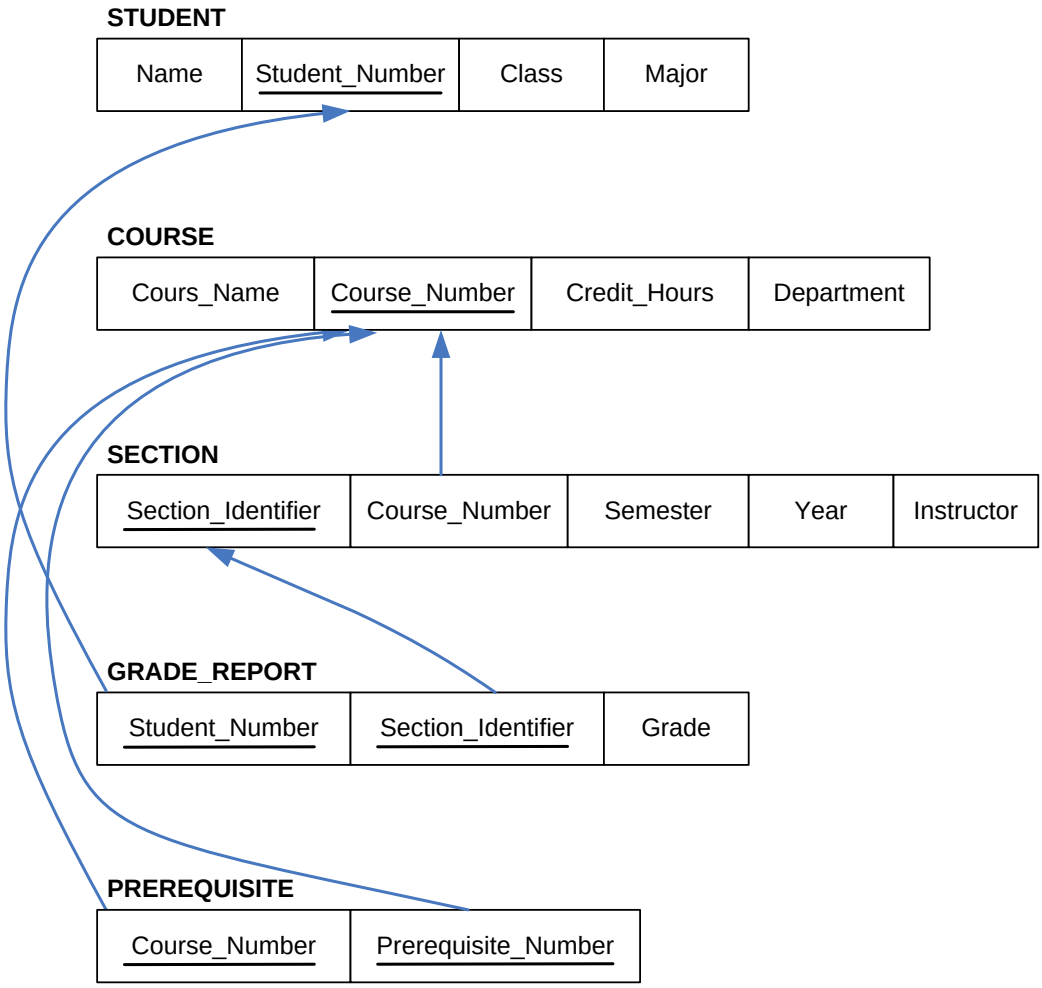
(a) Retrieve the names of all employees who work in the department that has the employee with the highest salary among all employees.
(b) Retrieve the names of all employees whose supervisor's supervisor has '888665555' for Ssn.
(c) Retrieve the names of employees who make at least $10,000 more than the employee who is paid the least in the company.

EXERCISE 4. (*Fundamentals of Database Systems*, Elmasri, Navathe, Addison-Wesley)

Specify the following queries in SQL on the schema of the following database.

(a) Retrieve the names of all senior students majoring in 'CS' (computer science).
(b) Retrieve the names of all courses taught by professor King in 2004 and 2005
(c) For each section taught by Professor King, retrieve the course number, semester, year, and number of students who took the course.
(d) Retrieve the name and transcript of each senior student (Class=4) majoring in CS. A transcript includes course name, course number, credit hours, semester, year, and grade for each course completed by the student.
(e) Retrieve the names and major departments of all straight-A students (students who have a grade of A in all their courses).
(f) Retrieve the names and major departments of all students who do not have any grade of A in any of their courses.

The Relational Schema of the University Database



STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

EXERCISE 4. *(from the book of Ramakrishnan and Gehrke)*

Consider the following schema:

SUPPLIERS (SID : *integer*, SNAME : *string*, ADDRESS : *string*)
PARTS (PID : *integer*, PNAME : *string*, COLOR : *string*)
CATALOG (SID : *integer*, PID : *integer*, COST : *real*)

The key fields are underlined, and the domain of each field is listed after the field name. Thus, SID is the key for SUPPLIERS, PID is the key for PARTS, and SID and PID together form the key for CATALOG. The CATALOG relation lists the prices charged for parts by suppliers. CATALOG.SID is a foreign key referring to SUPPLIERS.SID and CATALOG.PID is a foreign key referring to PARTS.PID.
Write the following queries in SQL.

1. Find the PNAMEs of parts for which there is some supplier.
2. Find the SNAMEs of suppliers who supply every part.
3. Find the SNAMEs of suppliers who supply every red part.
4. Find the PNAMESs of parts supplied by Acme Widget Suppliers and by no one else.
5. Find the SIDs of suppliers who charge more for some part than the average cost of that part (averaged over all the suppliers who supply that part).
6. For each part, find the SNAME of the supplier who charges the most for that part.
7. Find the SIDs of suppliers who supply only red parts.
8. Find the SIDs of suppliers who supply a red part and a green part.
9. Find the SIDs of suppliers who supply a red part or a green part.
10. Find the SIDs of suppliers who supply every part which is red or green.
11. Find the SIDs of suppliers who supply every red part or supply every green part.
12. For every supplier that supplies a green part and a red part, print her SID and the PID, the name and the price of the most expensive parts that she supplies.
13. List the PID, PNAME and average cost of all parts.
14. Find the average cost of parts supplied by suppliers named 'Yosemite Sham'.